

Titration report

LabX

Method

Method ID	CDC_GranTED-DYN	Release state	Released
Name	Titr TRIS with HCl - Gran Verification	Modified by	PredefinedUser
Modified on	2/5/2026 7:08:24 AM		

Summary

Samples

	Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.025853 mL	OK
		Concentration R2	0.10052 mol/L	OK
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.050831 mL	OK
		Concentration R2	0.10102 mol/L	OK

Statistics

R1

Mean	5.038342 mL	Minimum	5.025853 mL
Standard deviation	0.017662 mL	Rel. standard deviation	0.351 %
Total samples	2	Samples excluded	0

R2

Mean	0.10077 mol/L	Minimum	0.10052 mol/L
Standard deviation	0.00035 mol/L	Rel. standard deviation	0.351 %
Total samples	2	Samples excluded	0

Main Results

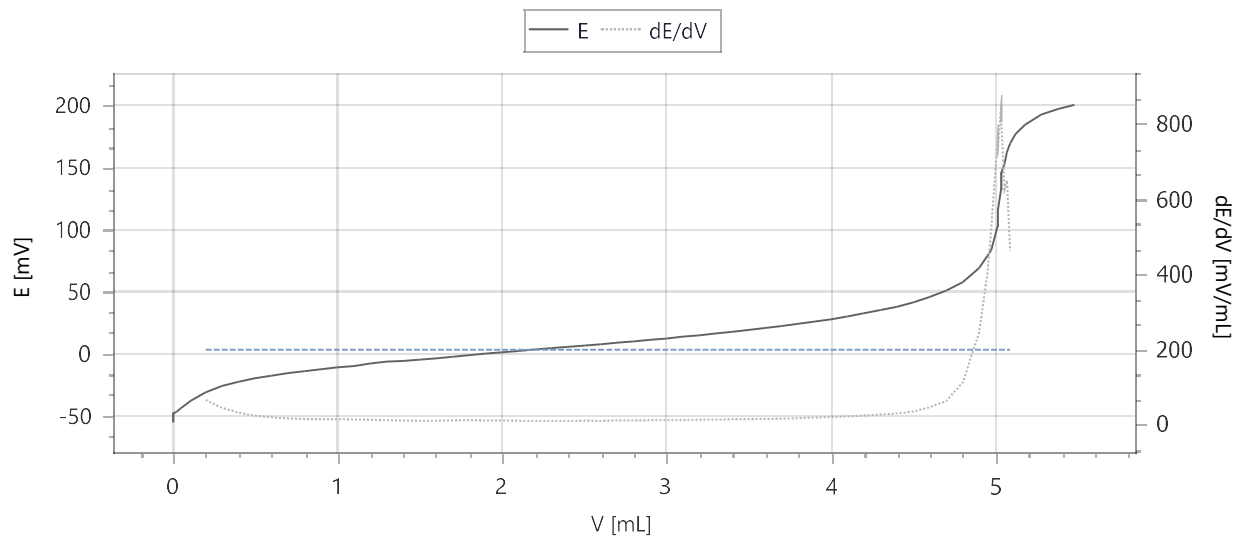
Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.025853 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10052 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.050831 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10102 mol/L	OK

Titration (EQP) [1] Scope 1/1, Sample 1/2

E - V & dE/dV - V Curve Scope 1/1, Sample 1/2

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Measured Values Scope 1/1, Sample 1/2

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-54.4		25.0			
34	0.0100	-48.5	0.0100	25.0	5.9		
37	0.0200	-47.2	0.0100	25.0	1.3		
41	0.0450	-44.3	0.0250	25.0	2.9		
44	0.1075	-38.0	0.0625	25.0	6.3		
48	0.1975	-31.3	0.0900	25.0	6.7	66.11	
53	0.2975	-26.3	0.1000	25.0	5.0	44.75	
59	0.3975	-22.8	0.1000	25.0	3.5	32.07	
64	0.4975	-20.0	0.1000	25.0	2.8	24.09	
70	0.5975	-17.9	0.1000	25.0	2.1	19.97	
74	0.6975	-15.7	0.1000	25.0	2.2	16.96	
79	0.7975	-14.2	0.1000	25.0	1.5	15.43	
84	0.8975	-12.6	0.1000	25.0	1.6	15.13	
89	0.9975	-11.2	0.1000	25.0	1.4	14.90	
94	1.0975	-10.2	0.1000	25.0	1.0	14.23	
98	1.1975	-8.0	0.1000	25.0	2.2	13.45	
101	1.2975	-6.6	0.1000	25.0	1.4	12.23	
105	1.3975	-5.9	0.1000	25.0	0.7	11.41	
110	1.4975	-4.9	0.1000	25.0	1.0	10.90	
114	1.5975	-3.8	0.1000	25.0	1.1	10.88	
118	1.6975	-2.5	0.1000	25.0	1.3	11.84	
122	1.7975	-1.2	0.1000	25.0	1.3	12.20	
124	1.8975	0.0	0.1000	25.0	1.2	11.72	
128	1.9975	1.1	0.1000	25.0	1.1	11.40	
131	2.0975	2.0	0.1000	25.0	0.9	10.99	
134	2.1975	3.3	0.1000	25.0	1.3	10.62	
138	2.2975	4.4	0.1000	25.0	1.1	10.85	
141	2.3975	5.5	0.1000	25.0	1.1	10.88	
145	2.4975	6.4	0.1000	25.0	0.9	10.99	
148	2.5975	7.5	0.1000	25.0	1.1	10.89	
151	2.6975	8.9	0.1000	25.0	1.4	11.39	

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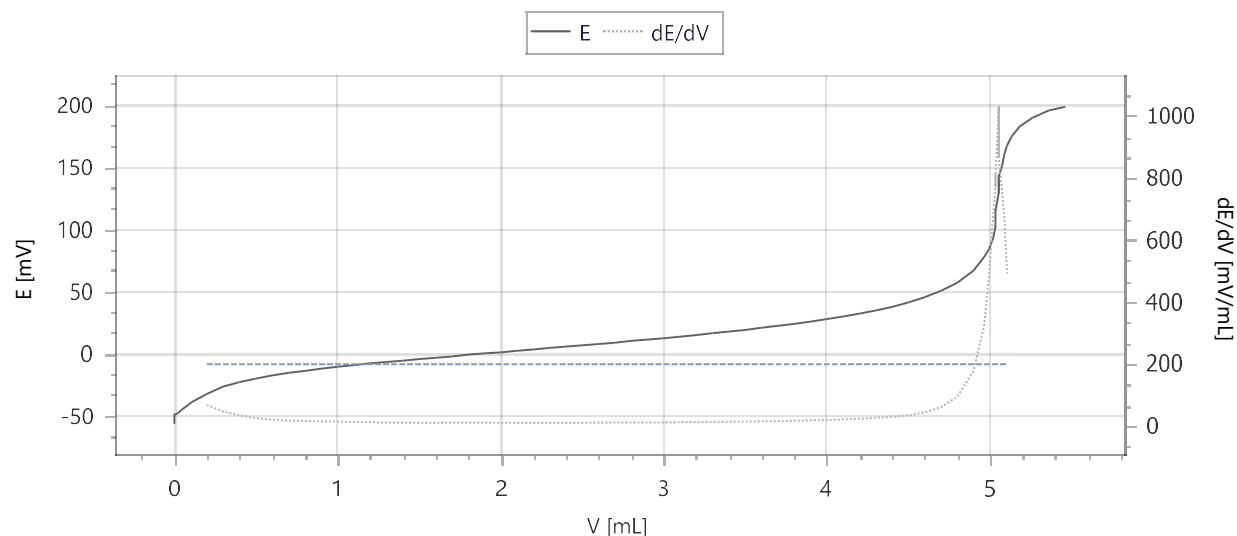
156	2.7975	9.8	0.1000	25.0	0.9	11.73	
159	2.8975	11.1	0.1000	25.0	1.3	12.06	
163	2.9975	12.2	0.1000	25.0	1.1	12.18	
166	3.0975	13.6	0.1000	25.0	1.4	12.67	
170	3.1975	14.8	0.1000	25.0	1.2	13.67	
174	3.2975	16.2	0.1000	25.0	1.4	14.03	
177	3.3975	17.6	0.1000	25.0	1.4	14.65	
180	3.4975	19.2	0.1000	25.0	1.6	15.14	
183	3.5975	20.8	0.1000	25.0	1.6	15.71	
187	3.6975	22.2	0.1000	25.0	1.4	16.29	
190	3.7975	24.0	0.1000	25.0	1.8	17.47	
193	3.8975	25.8	0.1000	25.0	1.8	19.15	
196	3.9975	27.7	0.1000	25.0	1.9	20.88	
200	4.0975	30.0	0.1000	25.0	2.3	22.51	
203	4.1975	32.5	0.1000	25.0	2.5	24.43	
208	4.2975	35.1	0.1000	25.0	2.6	27.38	
211	4.3975	37.8	0.1000	25.0	2.7	30.12	
214	4.4975	41.3	0.1000	25.0	3.5	35.20	
217	4.5975	45.7	0.1000	25.0	4.4	46.96	
222	4.6975	51.0	0.1000	25.0	5.3	63.48	
226	4.7975	57.8	0.1000	25.0	6.8	112.38	
234	4.8975	69.1	0.1000	25.0	11.3	246.20	
244	4.9475	79.0	0.0500	25.0	9.9	401.63	
249	4.9705	83.5	0.0230	25.0	4.5	519.49	
264	5.0120	103.7	0.0415	25.0	20.2	773.41	
278	5.0220	116.2	0.0100	25.0	12.5	741.31	
	5.025853	122.7				862.20	EQP
291	5.0320	133.0	0.0100	25.0	16.8	851.55	
298	5.0420	145.7	0.0100	25.0	12.7	808.10	
302	5.0520	153.9	0.0100	25.0	8.2	627.82	
305	5.0670	162.9	0.0150	25.0	9.0	650.59	
309	5.0855	169.8	0.0185	25.0	6.9	463.10	
313	5.1195	177.6	0.0340	25.0	7.8		
319	5.1770	185.0	0.0575	25.0	7.4		
322	5.2770	193.3	0.1000	25.0	8.3		
327	5.3770	197.7	0.1000	25.0	4.4		
330	5.4770	201.1	0.1000	25.0	3.4		

Titration (EQP) [1] Scope 1/1, Sample 2/2

E - V & dE/dV - V Curve Scope 1/1, Sample 2/2

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Measured Values Scope 1/1, Sample 2/2

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-55.8		25.0			
33	0.0100	-49.3	0.0100	25.0	6.5		
36	0.0200	-48.1	0.0100	25.0	1.2		
40	0.0450	-45.3	0.0250	25.0	2.8		
44	0.1075	-39.1	0.0625	25.0	6.2		
47	0.1985	-32.6	0.0910	25.0	6.5	68.06	
51	0.2985	-26.8	0.1000	25.0	5.8	47.34	
56	0.3985	-23.1	0.1000	25.0	3.7	34.43	
62	0.4985	-20.2	0.1000	25.0	2.9	26.08	
66	0.5985	-17.8	0.1000	25.0	2.4	21.15	
70	0.6985	-15.8	0.1000	25.0	2.0	17.94	
74	0.7985	-14.1	0.1000	25.0	1.7	16.68	
78	0.8985	-12.3	0.1000	25.0	1.8	15.60	
83	0.9985	-10.9	0.1000	25.0	1.4	14.63	
87	1.0985	-9.6	0.1000	25.0	1.3	13.68	
91	1.1985	-8.0	0.1000	25.0	1.6	12.69	
95	1.2985	-6.8	0.1000	25.0	1.2	11.62	
99	1.3985	-5.8	0.1000	25.0	1.0	11.33	
103	1.4985	-4.7	0.1000	25.0	1.1	11.15	
107	1.5985	-3.6	0.1000	25.0	1.1	11.10	
111	1.6985	-2.5	0.1000	25.0	1.1	11.35	
114	1.7985	-1.1	0.1000	25.0	1.4	11.37	
119	1.8985	-0.1	0.1000	25.0	1.0	11.25	
123	1.9985	0.9	0.1000	25.0	1.0	11.05	
127	2.0985	2.0	0.1000	25.0	1.1	10.85	
131	2.1985	3.2	0.1000	25.0	1.2	10.78	
134	2.2985	4.3	0.1000	25.0	1.1	10.83	
138	2.3985	5.3	0.1000	25.0	1.0	10.96	
142	2.4985	6.4	0.1000	25.0	1.1	11.05	
146	2.5985	7.5	0.1000	25.0	1.1	11.28	
150	2.6985	8.6	0.1000	25.0	1.1	11.50	

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153	2.7985	10.0	0.1000	25.0	1.4	11.76	EQP
157	2.8985	11.0	0.1000	25.0	1.0	11.93	
160	2.9985	12.2	0.1000	25.0	1.2	12.25	
164	3.0985	13.4	0.1000	25.0	1.2	12.54	
168	3.1985	14.8	0.1000	25.0	1.4	13.02	
171	3.2985	16.1	0.1000	25.0	1.3	13.80	
174	3.3985	17.5	0.1000	25.0	1.4	14.29	
178	3.4985	18.9	0.1000	25.0	1.4	14.94	
182	3.5985	20.5	0.1000	25.0	1.6	15.57	
186	3.6985	22.1	0.1000	25.0	1.6	16.52	
190	3.7985	23.8	0.1000	25.0	1.7	17.51	
194	3.8985	25.6	0.1000	25.0	1.8	18.87	
198	3.9985	27.5	0.1000	25.0	1.9	20.06	
201	4.0985	29.7	0.1000	25.0	2.2	21.79	
205	4.1985	32.0	0.1000	25.0	2.3	23.99	
209	4.2985	34.6	0.1000	25.0	2.6	26.95	
212	4.3985	37.3	0.1000	25.0	2.7	30.19	
216	4.4985	40.9	0.1000	25.0	3.6	34.86	
220	4.5985	45.0	0.1000	25.0	4.1	44.48	
225	4.6985	50.1	0.1000	25.0	5.1	60.91	
229	4.7985	56.6	0.1000	25.0	6.5	94.47	
235	4.8985	66.5	0.1000	25.0	9.9	177.57	
240	4.9635	77.1	0.0650	25.0	10.6	322.23	
243	4.9935	83.9	0.0300	25.0	6.8	480.56	
247	5.0190	92.7	0.0255	25.0	8.8	677.11	
250	5.0340	101.7	0.0150	25.0	9.0	776.29	
265	5.0440	114.6	0.0100	25.0	12.9	812.95	
	5.050831	125.1				1030.84	
279	5.0540	130.0	0.0100	25.0	15.4	1029.23	
286	5.0640	142.5	0.0100	25.0	12.5	865.53	
290	5.0740	151.0	0.0100	25.0	8.5	757.89	
294	5.0880	159.9	0.0140	25.0	8.9	671.62	
297	5.1045	166.9	0.0165	25.0	7.0	492.97	
300	5.1330	174.3	0.0285	25.0	7.4		
304	5.1830	182.4	0.0500	25.0	8.1		
309	5.2610	189.3	0.0780	25.0	6.9		
313	5.3610	195.2	0.1000	25.0	5.9		
319	5.4610	198.0	0.1000	25.0	2.8		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1
Sample ID 2

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Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
05.02.2026 07:17:44	PredefinedUser	Administrator